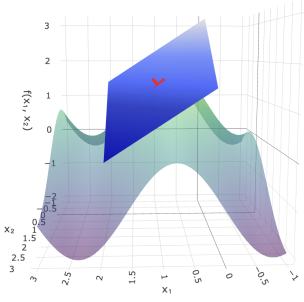
# Optimization in Machine Learning

## Mathematical Concepts
## Taylor Approximation

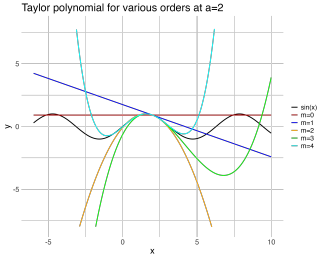


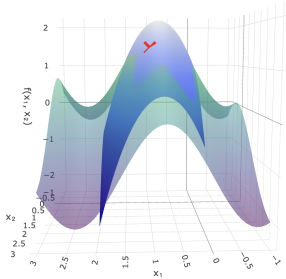

**Learning goals**

- Taylor's theorem (univariate)
- Taylor series (univariate)
- Taylor's theorem (multivariate)
- Taylor series (multivariate)

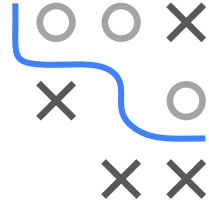
# TAYLOR APPROXIMATIONS: OVERVIEW

- To optimize (find minima and maxima) it can be extremely helpful to approximate nonlinear functions locally
- We can use **Taylor polynomials** to approximate functions and
- **Taylor's theorem** provides us with the tools to estimate the error of this approximation $\implies$ helpful for analyzing optimization algorithms
- Some functions can locally or even globally equal their **Taylor series**, i.e. the limit of Taylor polynomials



# TAYLOR APPROXIMATIONS: MOTIVATION



- Since the geometry of linear and quadratic functions is very well understood we will often want to use those for approximations
- For example, for a function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$

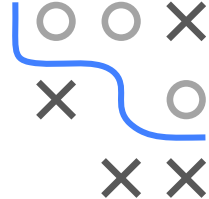
$$f(\boldsymbol{x} + \mathbf{h}) = f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h} + o(\mathbf{h})$$

- You might also often see an approximation via the gradient and Hessian of a function:

$$f(\boldsymbol{x} + \mathbf{h}) \approx f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h} + \frac{1}{2}\mathbf{h}^\top \nabla^2_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h}$$

- In fact, $f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h}$ and $f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h} + \frac{1}{2}\mathbf{h}^\top \nabla^2_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h}$ are, respectively, the first and second **Taylor polynomial of $f$ at $\boldsymbol{x}$**, evaluated at $\boldsymbol{x} + \mathbf{h}$

# TAYLOR POLYNOMIALS



- Idea: Find a polynomial that locally behaves like a function $f$ at point $\boldsymbol{a}$, i.e. matches $f$'s value ($f$), slope ($f'$), curvature ($f''$), etc.

$\Leftrightarrow$ Find polynomial so that

$$f(x) \approx T_k(x, \boldsymbol{a}) \quad \text{for all } x \text{ near } \boldsymbol{a}$$

where $k$ denotes the highest order of derivative of $f$ used in $T_k$

- Wording: We "*expand $f$ (via Taylor) around $\boldsymbol{a}$*"

**Definition of Taylor polynomial (univariate):** Let $I \subseteq \mathbb{R}$ be an open interval and $f \in \mathcal{C}^k(I, \mathbb{R})$. For each $a, x \in I$, the $k$th order Taylor polynomial for $f$ at $a$ is defined as

$$T_k(x, a) := \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!}(x-a)^j$$

# MULTIVARIATE TAYLOR POLYNOMIALS

For the multivariate version, we need a concise way to express derivatives and powers involving several variables

- A **multi-index** is a vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_d) \in \mathbb{N}^d$.
- Its **order** is the sum of its components: $|\boldsymbol{\alpha}| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$.
- Partial derivative is written as $D^{\boldsymbol{\alpha}} f = \dfrac{\partial^{|\boldsymbol{\alpha}|} f}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}$
- Factorials generalize componentwise: $\boldsymbol{\alpha}! = \alpha_1! \, \alpha_2! \cdots \alpha_d!$.
- For $\boldsymbol{x}, \boldsymbol{a} \in \mathbb{R}^d$: $(\boldsymbol{x} - \boldsymbol{a})^{\boldsymbol{\alpha}} = (x_1 - a_1)^{\alpha_1} \cdots (x_d - a_d)^{\alpha_d}$.

**Definition of Taylor polynomial (multivariate):** Let $I$ be an open subset of $\mathbb{R}^n$ and $f \in \mathcal{C}^k(I, \mathbb{R})$. For each $\boldsymbol{a}, \boldsymbol{x} \in I$, the $k$th order Taylor polynomial for $f$ at $\boldsymbol{a}$ is defined as

$$T_k(\boldsymbol{x}, \boldsymbol{a}) := \sum_{|\boldsymbol{\alpha}| \leq k} \frac{D^{\boldsymbol{\alpha}} f(\boldsymbol{a})}{\boldsymbol{\alpha}!} (\boldsymbol{x} - \boldsymbol{a})^{\boldsymbol{\alpha}}$$

# MULTIVARIATE TAYLOR POLYNOMIAL IDENTITIES

For $f \in \mathcal{C}^k(I, \mathbb{R})$ as before, we will often use the following identities:

- $T_1(\boldsymbol{x}, \boldsymbol{a}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a})$
- $T_2(\boldsymbol{x}, \boldsymbol{a}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a}) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{a})^T \boldsymbol{H}(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a})$

(Which squares with the notation of the motivation slide by setting $\boldsymbol{a} \hat{=} \boldsymbol{x}$ and $\boldsymbol{x} \hat{=} \boldsymbol{x} + \boldsymbol{h}$)

**Example: Specifically deriving $T_1$ for bivariate $f$ ($d = 2$)**

| $\alpha_1$ | $\alpha_2$ | $\lvert\boldsymbol{\alpha}\rvert$ | $D^{\boldsymbol{\alpha}} f$ | $\boldsymbol{\alpha}!$ | $(\boldsymbol{x} - \boldsymbol{a})^{\boldsymbol{\alpha}}$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | $f$ | 1 | 1 |
| 1 | 0 | 1 | $\partial f / \partial x_1$ | 1 | $x_1 - a_1$ |
| 0 | 1 | 1 | $\partial f / \partial x_2$ | 1 | $x_2 - a_2$ |

and, therefore

$$T_1(\boldsymbol{x}, \boldsymbol{a}) = \frac{f(\boldsymbol{a})}{1} \cdot 1 + \frac{\partial f(\boldsymbol{a})}{\partial x_1}(x_1 - a_1) + \frac{\partial f(\boldsymbol{a})}{\partial x_2}(x_2 - a_2)$$

$$= f(\boldsymbol{a}) + \begin{pmatrix} \frac{\partial f(\boldsymbol{a})}{\partial x_1} \\ \frac{\partial f(\boldsymbol{a})}{\partial x_2} \end{pmatrix}^T \begin{pmatrix} x_1 - a_1 \\ x_2 - a_2 \end{pmatrix} = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a})$$

# TAYLOR'S THEOREM

**General version for both univariate and multivariate functions:**
Let $I$ be an open subset of $\mathbb{R}^n$, $n \in \mathbb{N}_{>0}$, and $f \in \mathcal{C}^k(I, \mathbb{R})$. There exists a function $R_k : I \times I \to \mathbb{R}$ so that for each $\boldsymbol{a}, \boldsymbol{x} \in I$

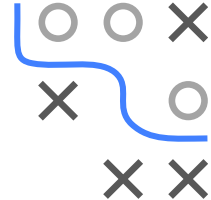
$$R_k(\boldsymbol{x}, \boldsymbol{a}) = o(\|\boldsymbol{x} - \boldsymbol{a}\|^k) \text{ as } \boldsymbol{x} \to \boldsymbol{a}$$

and

$$f(\boldsymbol{x}) = T_k(\boldsymbol{x}, \boldsymbol{a}) + R_k(\boldsymbol{x}, \boldsymbol{a})$$

- $R_k(\boldsymbol{x}, \boldsymbol{a})$ is called **remainder term** and different specific forms have been established
- However, we will usually focus on the property $R_k(\boldsymbol{x}, \boldsymbol{a}) = o(\|\boldsymbol{x} - \boldsymbol{a}\|^k)$ as $\boldsymbol{x} \to \boldsymbol{a}$ or upper bounds derived for specific function classes when analyzing optimization algorithms

# TAYLOR SERIES

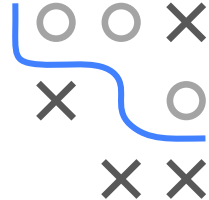


- For $f \in \mathcal{C}^\infty$, there might exist an open ball $B_r(\boldsymbol{a})$ with radius $r > 0$ around $\boldsymbol{a}$ such that the **Taylor series**

$$T_\infty(\boldsymbol{x}, \boldsymbol{a}) = \begin{cases} \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k & \text{if } f \text{ is univariate} \\ \sum_{|\alpha| \geq 0} \frac{D^\alpha f(\boldsymbol{a})}{\alpha!}(\boldsymbol{x}-\boldsymbol{a})^\alpha & \text{if } f \text{ is multivariate} \end{cases}$$

  converges to $f$ on $B_r(\boldsymbol{a})$
- If such an open Ball exists for all $\boldsymbol{a}$ in the domain of $f$, $f$ is called an **analytic function**
- Even if Taylor series converges, it might not converge to $f$
- Upper bound $R = \sup\{r \mid \text{Taylor series converges on } B_r(\boldsymbol{a})\}$ is called the radius of convergence of Taylor series around $\boldsymbol{a}$
- If $R > 0$ and $f$ analytic, Taylor series converges absolutely and uniformly to $f$ on compact sets inside $B_R(\boldsymbol{a})$
- No general convergence behaviour on boundary of $B_R(\boldsymbol{a})$

# EXAMPLES OF ANALYTIC FUNCTIONS



For analytic functions the remainder term eventually vanishes, i.e. $R_k(\boldsymbol{x}, \boldsymbol{a}) \to 0$ as $k \to \infty$ for all $\boldsymbol{x} \in B_r(\boldsymbol{a})$.

Important examples are

- Polynomials
- Exponential function (exp)
- Trigonometric functions (sin, cos)

And important rules are

- Any analytic function of a polynomial is again an analytic function
- Analytic functions are closed under sum and product (due to the properties of series)
- The derivative of an analytic function is again an analytic function

**One specific example:** $f : \mathbb{R}^2 \longrightarrow \mathbb{R}\ \boldsymbol{x} \mapsto \sin(2x_1) + \cos(x_2)$

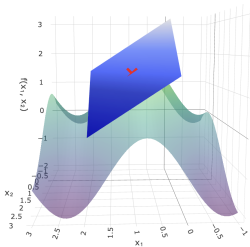
# EXAMPLE: TAYLOR APPROXIMATION OF $f(\boldsymbol{x}) = \sin(2x_1) + \cos(x_2)$ AT $\boldsymbol{a} = (1,1)^T$

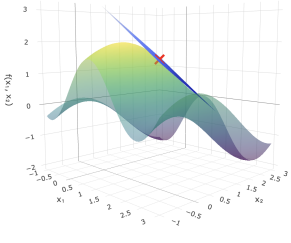
**1st order**: we know that $f(\boldsymbol{x}) = \underbrace{f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a})}_{T_1(\boldsymbol{x},\boldsymbol{a})} + R_1(\boldsymbol{x}, \boldsymbol{a})$ and since

$\nabla f(\boldsymbol{x}) = (2\cos(2x_1), -\sin(x_2))$,

$$f(\boldsymbol{x}) = T_1(\boldsymbol{x}) + R_1(\boldsymbol{x}, \boldsymbol{a}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a}) + R_1(\boldsymbol{x}, \boldsymbol{a})$$

$$= \sin(2) + \cos(1) + (2\cos(2), -\sin(1)) \begin{pmatrix} x_1 - 1 \\ x_2 - 1 \end{pmatrix} + R_1(\boldsymbol{x}, \boldsymbol{a})$$

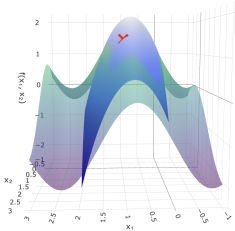
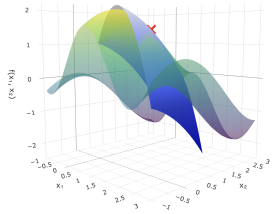


# EXAMPLE: TAYLOR APPROXIMATION OF $f(\boldsymbol{x}) = \sin(2x_1) + \cos(x_2)$ AT $\boldsymbol{a} = (1,1)^T$

**2nd order**: we know that

$$f(\boldsymbol{x}) = \underbrace{f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x}-\boldsymbol{a}) + \frac{1}{2}(\boldsymbol{x}-\boldsymbol{a})^T \boldsymbol{H}(\boldsymbol{a})(\boldsymbol{x}-\boldsymbol{a})}_{T_2(\boldsymbol{x},\boldsymbol{a})} + R_2(\boldsymbol{x},\boldsymbol{a})$$

and since $H(\boldsymbol{x}) = \begin{pmatrix} -4\sin(2x_1) & 0 \\ 0 & -\cos(x_2) \end{pmatrix}$,

$$f(\boldsymbol{x}) = T_1(\boldsymbol{x},\boldsymbol{a}) + \frac{1}{2}\begin{pmatrix} x_1 - 1 \\ x_2 - 1 \end{pmatrix}^T \begin{pmatrix} -4\sin(2) & 0 \\ 0 & -\cos(1) \end{pmatrix} \begin{pmatrix} x_1 - 1 \\ x_2 - 1 \end{pmatrix} + R_2(\boldsymbol{x},\boldsymbol{a})$$